

Tameem Tantawy

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EDUCATION

University of South Florida

Tampa, FL

Bachelors of science in Computer Science - GPA: 3.75/4.0

Expected Graduation: May 2027

Relevant Courses: Operating Systems, Data Structures, Algorithms (Princeton University), Database Design, Programming Concepts (Python), Program Design (C), Computer Organization, Discrete Structures, Linear Algebra

TECHNICAL SKILLS

Programming Languages: Rust, Python, Scala, Java, JavaScript, TypeScript, Go, SQL, HTML, CSS

Technologies: Apache Spark, Apache Airflow, Apache Kafka, Apache Pinot, gRPC, Docker, Linux, Grafana, AWS, Git, CI/CD

Databases & Frameworks: PostgreSQL, MongoDB, Django, FastAPI, React

EXPERIENCE

LinkedIn

Mountain View, CA

Software Engineering Intern - (Java, Scala, Spark)

May 2026 – August 2026

- Built distributed Scala/Apache Spark data pipelines powering multiple LinkedIn Premium Company Page systems across **23M+ Company Pages**, processing large-scale datasets through Apache Airflow-orchestrated workflows for reliable downstream serving.
- Reduced pipeline runtime by 75%, from 8 hours to ~2 hours** by parallelizing independent signal computations through a **Scala thread pool**, improving concurrency and throughput while engineering fault-tolerant workflows for reliable production execution.
- Architected the storage and low-latency serving architecture for precomputed signals, evaluating **Pinot vs. Venice** across consistency, availability, latency, and scalability tradeoffs, and built Java backend services using gRPC for low-latency retrieval and downstream propagation.

Amazon Web Services

Redmond, WA

Software Engineering Intern - (Rust, Caching, Memory Management)

September 2025 – December 2025

- Designed and implemented a Rust cache layer for Hermes**, a distributed caching system accelerating Apache Spark queries over Amazon S3 and Apache Iceberg, reducing millions of repeated remote metadata and schema lookups.
- Designed and implemented a ghost-buffer data structure to predict cache hit-rate improvements, using a multi-layer design for the DRAM layer and a sharded design for the disk layer to **maintain correctness and handle high concurrency in a distributed cache**.
- Enabled cache-capacity experimentation without allocating additional cache memory, **reducing memory overhead by 90% while predicting post-resize cache hit rates within 1–2% of observed results**.

Dropbox

San Francisco, CA

Software Engineering Intern - (Rust, Python, gRPC)

May 2025 – August 2025

- Architected Python and Rust backend infrastructure using gRPC to propagate configuration state across Dropbox's desktop app and Nucleus sync engine, **supporting 80M+ DAU with consistent state replication across platforms**.
- Implemented observability with Grafana dashboards, and added integration tests and Unit Tests, providing monitoring of adoption, error rates, and latency to validate correctness and reliability at global scale, adhering with Dropbox's goal of achieving **99.999% availability**.
- Improved fault tolerance in Dropbox's desktop configuration path** by adding validation, fallback behavior, and failure-case integration coverage across the app, local configuration layer, and Nucleus sync engine.

SHIELD Lab at USF

Tampa, FL

Undergraduate Research Assistant - (Python, Web Sockets)

September 2024 – May 2025

- Developed an app for the Mac Parkman Foundation, aiming to reach over **5,000 individuals** and raise awareness about concussion prevention through engaging educational outreach, featuring live 3D animations, interactive graphs, and real-time metrics.
- Built a scalable back-end with Django, leveraging Web Sockets to support up to **1,000** data points per second received from external sensors, ensuring seamless data flow and minimizing any delay to the front-end.
- Planned software development life cycles using Sprint methodologies, ensuring timely delivery of features through CI/CD pipelines.

PROJECTS

Jamaa

Tampa, FL

- Led a team of **3** to co-develop a cross-platform mobile app that connected users with their local Muslim community by enabling mosques to share posts, events, and prayer times, resulting in increased community engagement and participation.
- Utilized Django's REST Framework to develop **25+** RESTful APIs for different types of users and optimized data storage using table partitioning and indexing in a PostgreSQL database, achieving a **53%** reduction in query response time.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

ColorStack USF | Tech Lead

May 2024 - Present

- Recruited **50+** underrepresented students in STEM and promoted the ColorStack mission, increasing chapter visibility and engagement.
- Hosted a back-end development workshop, helping **20+** members to gain hands-on experience and improve their technical proficiency.

Affiliations: Figma Figfest(1 of 62), Google Cloud Career Jumpstart Participant(1 of 112 selected), Databricks BrickFest(1 of 50 selected), Salesforce Tech Accelerator(1 of 80 selected), IMC Trading Launchpad(1 of 30 selected), Citi EID Program (Technology Track), Accenture Innovation Challenge(1 of 150 selected)